

Problem

Calculate the line current required for a 30-kW three-phase motor having a power factor of 0.85 lagging if it is connected to a balanced source with a line voltage of 550 V.

Step-by-step solution

Step 1 of 2

The power factor can be expressed as the ratio of real power P to the apparent power S .

$$\text{pf} = \frac{P}{S}$$

Calculate the value of the apparent power S .

$$S = \frac{P}{\text{pf}}$$

Substitute 30 kW for P and 0.85 for pf.

$$\begin{aligned} S &= \frac{30 \times 10^3}{0.85} \\ &= 35.29 \text{ kVA} \end{aligned}$$

Therefore, for the value of the apparent power S is, 35.29 kVA.

Step 2 of 2

Consider the expression for the apparent power.

$$S = \sqrt{3} V_L I_L$$

Substitute 35.29 kW for S and 550 V for V_L .

$$\begin{aligned} 35.29 \text{ k} &= \sqrt{3} (550) I_L \\ I_L &= \frac{35.29 \text{ k}}{\sqrt{3} (550)} \\ &= 37.05 \text{ A} \end{aligned}$$

Therefore, the required value of the line current I_L is, 37.05 A.